

HS 224: Game Theory and Economics

BTech 4th Semester, Quiz 1

Total Marks 10, Time 55 minutes

Date: 6th Feb, 2018

[Answers should be accompanied by proper elaboration]

1. [Guessing two-thirds of the average game] Each of three people announces an integer from 1 to K . If the three integers are different, the person whose integer is closest to $2/3$ of the average of the three integers wins Rs.100. If two or more integers are the same, Rs.100 is split equally between the people whose integer is closest to $2/3$ of the average integer. Is there any integer k such that the action profile (k, k, k) , in which every person announces the same integer k , is a Nash equilibrium? (Hint: If $k \geq 2$, what happens if a person announces a smaller number?) [4] BD

2. [Fiscal policy-monetary policy game] The Central Government (G) and the Reserve Bank (RBI) face the following hypothetical situation. The government wants to spend more money to boost the GDP. But high spending policy results in budget deficit (BD), which leads to inflation. If G does not spend a lot the budget is balanced (BB), and there is no inflation. For the government BD is preferable to BB. On the other hand, the RBI can keep a high interest rate (HI) or lower the interest rate (LI). G wants the interest rate to be low, so much so that low interest in balanced budget is preferable to high interest in budget deficit. High interest policy is used by the RBI to control inflation. So, RBI prefers HI over LI in case of BD. Otherwise, low interest rate is preferable. For the RBI BB is more important, so that BD with low interest rate is worse than BB with high interest rate.

(a) Model the situation as a strategic game and find its Nash equilibrium/equilibria. Is the game similar to the Prisoner's Dilemma game in some way? [3+1]

(b) Is there any dominated action in this game? If yes, find the equilibrium through elimination of dominated actions. [2]

